

Handshake Cost and Size in PQC TLS 1.3

Abstract

Post-quantum TLS 1.3 changes the operational cost of a handshake. This manuscript measures handshake latency, CPU overhead, and transmitted bytes for classical, hybrid, and post-quantum configurations under controlled conditions.

Methodology

The benchmark fixes client, server, network, and cryptographic configuration before varying the key exchange and certificate profile. Each run records timing data, CPU measurements, handshake size, negotiated parameters, and failure conditions. The comparison is organized around reproducible command lines and stable report schemas.

Results

The current results make migration cost visible at the systems level: handshake-size growth, latency shifts, CPU overhead, and configuration-dependent tradeoffs across classical, hybrid, and post-quantum TLS 1.3.